RELATIVE PERFECTNESS CHECKED ON AFFINE OPENS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Being pseudo-coherent and having locally finite tor dimension are local properties. used at proof lines 8966-8968. Reduces the equivalence to the case where X and S are affine.
- On affines D_QCoh is the derived category of modules and E corresponds to K = RGamma(X, E). invoked at proof lines 8971-8973. Sets up the module-level comparison.
- By definition relative perfectness is pseudo-coherence plus finite tor dimension over the base. recalled at proof lines 8976-8979. Provides the two conditions to translate.
- On affines pseudo-coherence and finite tor dimension over the base of an object agree with those of its module complex. invoked at proof lines 8980-8986. Completes the affine equivalence, hence all three statements.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 8971-8973 to identify K = RGamma(X, E); the translation of the conditions is inside the counted span.

Source reference. Original definition 0DHS: sources/stacks-chow/tags.txt, source lines 16917–16917. Cited at proof lines 8978-8979 for the characterization of R-perfect complexes; the translations are inside the counted span.

Source reference. Original lemma 08E7: sources/stacks-perfect/pinned-original-source.tex, source lines 2440–2440. Cited at proof lines 8980-8982 to translate pseudo-coherence; the conclusion is inside the counted span.

Source reference. Original lemma 0DHY: sources/stacks-perfect/pinned-original-source.tex, source lines 2552–2552. Cited at proof lines 8983-8986 to translate finite tor dimension; the conclusion is inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes which is flat and locally of finite presentation. Let E be an object of $D_{QCoh}(\mathcal{O}_X)$. The following are equivalent*

(1) *E is S-perfect,*
(2) *for any affine open $U \subset X$ mapping into an affine open $V \subset S$ the complex $R\Gamma(U, E)$ is $\mathcal{O}_S(V)$-perfect.*

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(3) *there exists an affine open covering* $S = \bigcup V_i$ *and for each* i *an affine open covering* $f^{-1}(V_i) = \bigcup U_{ij}$ *such that the complex* $R\Gamma(U_{ij}, E)$ *is* $\mathcal{O}_S(V_i)$*-perfect.*

Proof. Being pseudo-coherent is a local property and "locally having finite tor dimension" is a local property. Hence this lemma immediately reduces to the statement: if X and S are affine, then E is S-perfect if and only if $K = R\Gamma(X, E)$ is $\mathcal{O}_S(S)$-perfect. Say $X = \mathrm{Spec}(A)$, $S = \mathrm{Spec}(R)$ and E corresponds to $K \in D(A)$, i.e., $K = R\Gamma(X, E)$, see Lemma 06Z0.

Observe that K is R-perfect if and only if K is pseudo-coherent and has finite tor dimension as a complex of R-modules (More on Algebra, Definition 0DHS). By Lemma 08E7 we see that E is pseudo-coherent if and only if K is pseudo-coherent. By Lemma 0DHY we see that E has finite tor dimension over $f^{-1}\mathcal{O}_S$ if and only if K has finite tor dimension as a complex of R-modules. □